

Welcome to RASCAL!

Desktop version - User Manual

Version 1.8



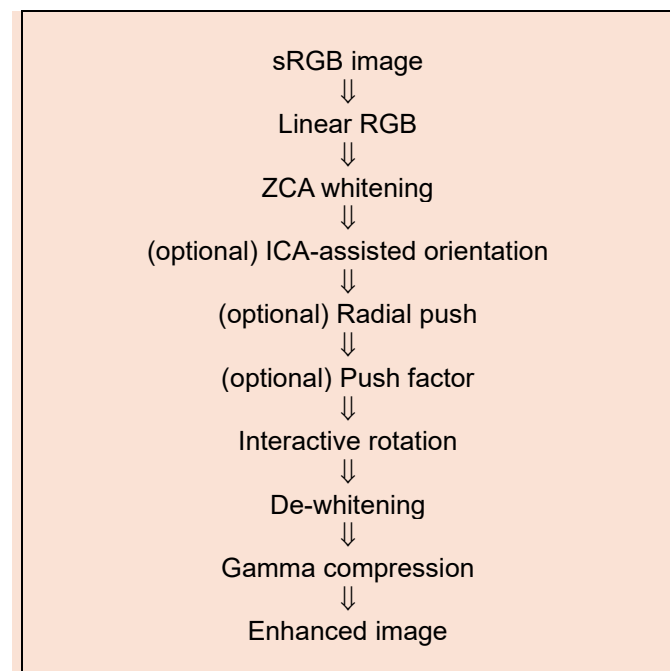
Edition	Desktop software
Platform	Windows 10+ / macOS Intel & Apple Silicon version / Linux Ubuntu

1. Presentation

RASCAL (**R**adial **A**ngular **S**tatistical **C**olor **A**daptation and **L**ifting) is an interactive software designed to reveal subtle chromatic variations in standard RGB digital images. The program was developed primarily for applications in cultural heritage and archaeological imaging, where faint color differences may correspond to pigments, surface alterations, or archaeological structures that are difficult to detect visually. The software performs a decorrelation of RGB color channels followed by an interactive exploration of color transformations. This approach allows the user to visually enhance color contrasts and to concentrate relevant chromatic information on one or several of the image channels. Unlike fully automated enhancement techniques, RASCAL relies on seamless interactive exploration. The operator visually evaluates the transformations and identifies the most informative configuration.

2. Conceptual overview

Digital images are typically encoded in the sRGB color space, where the three-color channels may be strongly correlated. Such correlations often confine color information to a restricted region of the RGB cube, which reduces the visibility of some chromatic variations. RASCAL addresses this limitation using the following workflow:



After whitening in the linear RGB color space, the covariance matrix of the color distribution becomes spherical, meaning that all chromatic directions have unit variance. Because the covariance structure is isotropic after whitening, any orthogonal rotation preserves distances and variances in the whitened space. The user can then rotate the color space to align chromatic variations of interest with the RGB axes. Note that the whitening transformation is computed once from the original image (or from a subset of it) and remains fixed during the entire exploration. ICA-assisted orientation can also be chosen as starting point, as it may help separate statistically independent chromatic components. In any case, subsequent rotations do not modify the statistical model. As a result, the rotation performed

by RASCAL does not correspond to a simple rotation of the RGB axes. Instead, the rotation operates in the whitened color space, whose geometry depends on the covariance structure of the image. Consequently, the transformation is image-dependent and reflects the internal geometry of the color distribution. This process often reveals structures that are difficult or impossible to perceive in the original image. After rotation, the transformed data are mapped back to the original RGB space through the inverse whitening (de-whitening) transform, restoring the original variance structure. The resulting linear RGB values are then converted back to sRGB through gamma compression for display.

The detailed mathematical description of the method is provided in the ‘Materials and Methods’ section of the associated article.

3. Technical overview

RASCAL is written in Python 3.10. The main libraries used are PyQt5, VisPy, PyOpenGL, NumPy, Pillow, and ImageIO. Interactive visualization and color-space rotation are GPU-accelerated through GLSL shaders. Some operations, such as Radial Push computation, image quality assessment, session saving/loading, and full-resolution exports, are computed on the CPU. The application is optimized for displays with a Full HD resolution (1920 × 1080 px) or higher.

4. Installation

Linux. Extract the RASCAL archive to the folder of your choice (tested on Ubuntu 26.04 LTS). Open a terminal in the download directory and make the file executable:

```
chmod +x Rascal-x86_64.AppImage
```

then run:

```
./Rascal-x86_64.AppImage
```

To run RASCAL **from source** instead, on Debian/Ubuntu-based systems:

```
sudo apt update
```

```
sudo apt install python3 python3-pip python3-venv libgl1 libglx-mesa0 libegl1  
libxkbcommon-x11-0 libxcb-xinerama0
```

If Qt/OpenGL problems persist, the following additional packages may also be required:

```
sudo apt install libxcb-icccm4 libxcb-keysyms1 libxcb-render-util0 libxcb-  
cursor0
```

Open a terminal in the RASCAL source folder and run:

```
python3 -m venv rascal_env  
source rascal_env/bin/activate  
python -m pip install --upgrade pip  
python -m pip install -r requirements.txt  
python Rascal.py
```

Each time you want to run RASCAL from the same folder:

```
source rascal_env/bin/activate
python Rascal.py
```

Troubleshooting:

Imagecodecs. This package improves support for some compressed TIFF files, especially LZW/Deflate-compressed TIFFs. On some systems, especially ARM or minimal distributions, the pre-built wheel may not be available. Install the build dependencies first:

```
sudo apt install liblzma-dev libjpeg-dev zlib1g-dev libtiff-dev build-essential

python -m pip install imagecodecs
```

If it still fails, RASCAL will still run without imagecodecs, but some compressed TIFF files may not open correctly.

Rawpy. This package is required only for camera RAW files such as NEF, CR2, ARW, RAF, ORF, RW2, and similar formats. On x86_64 systems, the pip wheel should usually install without issues. On ARM or other architectures:

```
sudo apt install libraw-dev

python -m pip install rawpy
```

If rawpy cannot be installed, RASCAL will still open standard image formats such as JPEG, PNG, BMP, and TIFF.

psutil. This package is used only for memory monitoring in the status bar. On most systems, it installs without issues. If it fails:

```
sudo apt install python3-dev build-essential

python -m pip install psutil
```

RASCAL will run normally without psutil; only the memory indicator will be unavailable.

PyQt5. On some recent distributions, for example Ubuntu 24.04 or later, the pip PyQt5 wheel may conflict with system Qt libraries. In that case, use the system package instead:

```
sudo apt install python3-pyqt5 python3-pyqt5.qtopengl
```

Windows. The Windows version of the software is distributed as an **installer file** (.msi). To install RASCAL, simply run the installer, preferably with administrator privileges, follow the standard installation steps, and then launch RASCAL from the Start Menu or from the desktop shortcut created during installation.

To run RASCAL from **source instead**, install Python 3.10 or later, then open a terminal in the RASCAL source folder and run:

```
python -m venv rascal_env
```

```
rascal_env\Scripts\activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
python Rascal.py
```

Each time you want to run RASCAL from the same folder:

```
rascal_env\Scripts\activate
python Rascal.py
```

macOS. The macOS version of the software is distributed as a **disk image** file (.dmg). To install the application, open the .dmg file and drag the RASCAL application into the Applications folder. Once installed, launch RASCAL from the Applications folder or via Spotlight. On first launch, macOS may display a security warning because the application is not distributed through the App Store. If this occurs, right-click the application, select Open, and then confirm. This operation is required only once.

To run RASCAL **from source** instead, install Python 3.10 or later from python.org or with Homebrew.

```
brew install python@3.10
```

Depending on your system configuration, the command may be python3, python3.10, or another Python executable. In the commands below, use the executable corresponding to your Python 3.10 or later installation.

Open a terminal in the RASCAL source folder and run:

```
python3 -m venv rascal_env
source rascal_env/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
python Rascal.py
```

Each time you want to run RASCAL from the same folder:

```
source rascal_env/bin/activate
python Rascal.py
```

Troubleshooting:

Imagecodecs. This package improves support for some compressed TIFF files, especially LZW/Deflate-compressed TIFFs. Pre-built wheels are generally available for common macOS architectures. If installation fails, first make sure that pip is up to date:

```
python -m pip install imagecodecs
```

If it still fails, RASCAL will still run without imagecodecs, but some compressed TIFF files may not open correctly.

Rawpy. This package is required only for camera RAW files such as NEF, CR2, ARW, RAF, ORF, RW2, and similar formats. If installation fails:

```
python -m pip install --upgrade pip
```

```
python -m pip install rawpy
```

Without rawpy, RASCAL will still open standard image formats such as JPEG, PNG, BMP, and TIFF.

psutil. This package is used only for memory monitoring in the status bar. If installation fails, install the Xcode command-line tools and try again:

```
xcode-select --install
```

```
python -m pip install psutil
```

RASCAL will run normally without psutil; only the memory indicator will be unavailable.

PyQt5. On older Macs, try setting `PyQt5>=5.12.3` in the requirements file. Recent PyQt5 wheels are available for both Intel Macs and Apple Silicon Macs. If PyQt5 installation nevertheless fails, try :

```
python -m pip install --only-binary=:all: -r requirements.txt
```

If problems persist on Apple Silicon, an alternative is to use Homebrew's PyQt5 package:

```
brew install pyqt@5
```

```
python3 -m venv --system-site-packages rascal_env
```

```
source rascal_env/bin/activate
```

```
python -m pip install -r requirements.txt
```

In that case, make sure that requirements.txt does not force a conflicting pip installation of PyQt5.

OpenGL warnings. macOS has deprecated OpenGL since Catalina 10.15. RASCAL may display OpenGL deprecation warnings in the terminal. These warnings are expected and can usually be ignored.

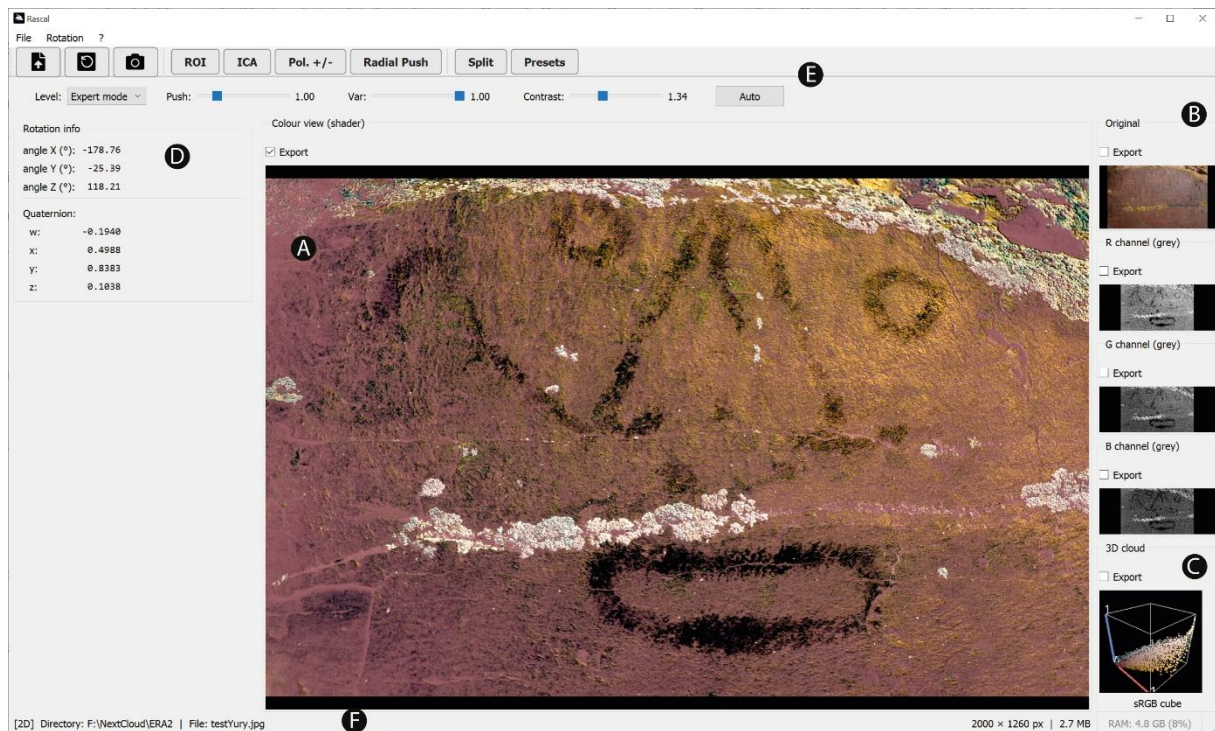


Figure 1: Screenshot of the layout for processing a single image (Expert mode). A: Main window; B: Thumbnails of the composite image and the three channels; C: 3D cloud for RGB cube or the whitened space; D: Information panel providing angles for rotation; E: Toolbar providing quick access to frequently used actions; F: Status bar displaying contextual information about the application's current state.

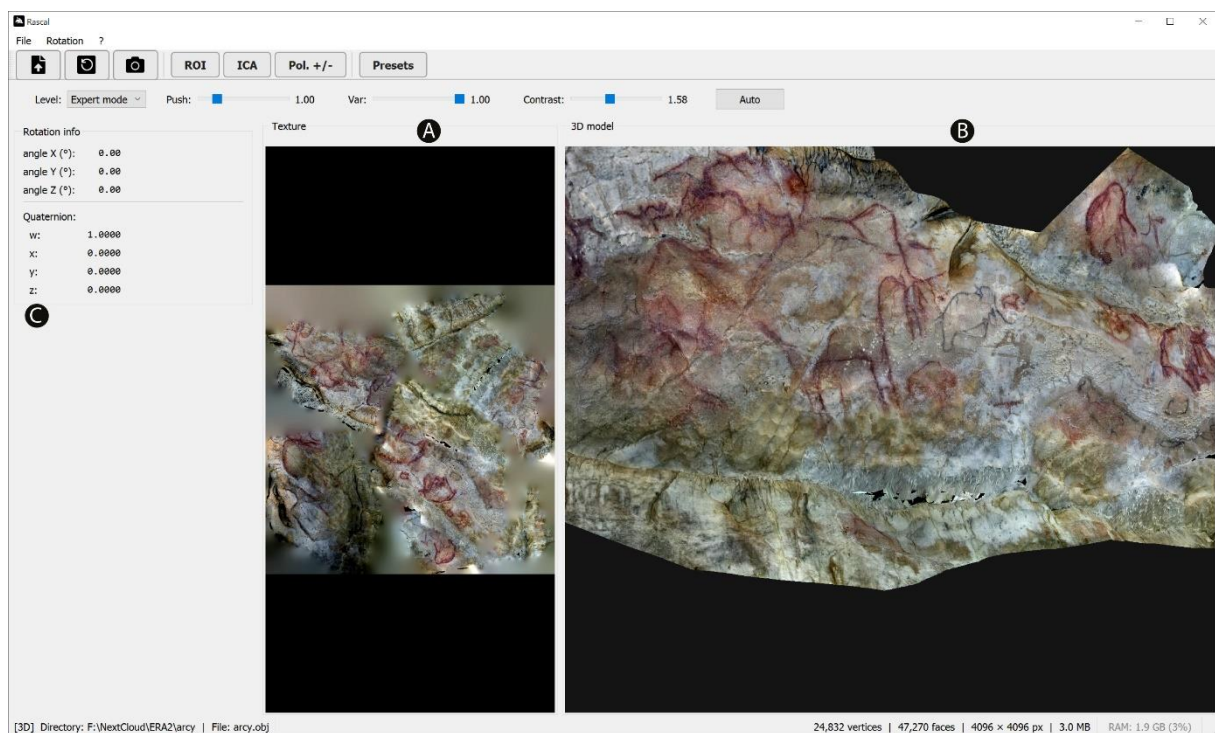


Figure 2: Screenshot of the layout for processing textured 3D models (Expert mode). A: Texture window; B: 3D model window; C: Information panel providing angles for rotation.

5. Interface description

Processing a single image. The interface can operate in two modes: Basic mode and Expert mode. Basic mode provides access to the main visualization tools and interactive rotation of the color space. Expert mode gives access to additional processing options such as Region-of-Interest (ROI), Independent Component Analysis (ICA), Radial Push, Push Factor, Variance restoration (Var) processing, as well as Presets and the information panel describing the current rotation. The mode can be switched at any time using the corresponding button in the interface. The interface is composed of several main elements (cf. Fig. 1):

- A. *Main image window* (in all modes). The central panel displays the transformed image after color processing. Changes in color-space rotation are immediately visible in this window.
- B. *Channel views* (in all modes). Three smaller panels display the Original, and individual R, G and B channels of the transformed image in greyscale. These latter views help identify which axis contains the most relevant information. The views displayed in the right panels can be swapped with the main image window either by clicking on them or by using the keyboard shortcuts **Ctrl + 1**, **Ctrl + 2**, and **Ctrl + 3** / **⌘ + 1**, **⌘ + 2**, and **⌘ + 3** on macOS. The Original view displays the unmodified source image for comparison. A single left click on the Original view swaps it with the current processed view.
- C. *3D cloud* (only in Expert Mode). A three-dimensional visualization shows the distribution of sampled pixel colors. Two display modes are available: (i) the RGB cube and (ii) the decorrelated (whitened) color space, optionally transformed by radial push. The view can be toggled using the **N** key. This visualization helps interpret the effect of color-space rotations. The cube itself can be rotated by dragging with the **left mouse button** on the view and zoomed in or out using the **right mouse drag** or the **mouse wheel**.
- D. *Information panel* (only in Expert mode). The current rotation may be displayed as numerical information expressed as Euler angles and quaternions.
- E. *Toolbar* provides quick access to frequently used actions.
- F. *Status bar* displays contextual information about the application's current state: file and directory, resolution, memory usage, etc.

Processing a textured 3D model. As in the single-image layout, the interface can operate in two modes: Basic mode and Expert mode. Basic mode provides access to the main visualization tools and interactive rotation of the color space. Expert mode gives access to additional processing options such as Region-of-Interest (ROI), Independent Component Analysis (ICA), Push Factor, Variance Restore, Presets, and the information panel. Radial Push and Split View are not available for textured 3D models. The interface is composed of several main elements (cf. Fig. 2):

- A. *Main texture image window* (in all modes). The left panel displays the transformed texture image after color processing. Changes in color-space rotation are immediately visible in this window.
- B. *Main textured 3D model window* (in all modes). Changes in color-space rotation performed in the main texture image window are immediately visible in this window.

- C. *Information panel* (only in Expert mode). The current rotation may be displayed as numerical information expressed as Euler angles and quaternions.

6. Starting the application

When the application starts, a welcome screen is displayed. From this screen, the user can open an image, a textured 3D model, or a previously saved .rasc session, either from the File menu or by drag and drop.

Loading an image. Images can be opened using:

File → Open (or the corresponding toolbar icon , or **Ctrl + O** / **⌘ + O** on macOS, or using drag and drop).

Supported file formats for images are JPEG, PNG, BMP, and TIFF (3×8-bit, 3×12-bit and 3×16-bit, LZW-compressed or uncompressed), and RAW files from most common camera manufacturers (Nikon, Canon, Sony, Fujifilm, Olympus, Panasonic, Pentax, etc). High bit depths are thus supported but usually provide no practical advantage while increasing memory usage. For best results, high-quality images should be used. TIFF images converted from RAW camera files often provide optimal results. If an image contains an alpha channel, only the RGB channels are used. Grayscale images can be loaded, but they are internally converted to three identical RGB channels. Since RASCAL is designed to explore chromatic differences, results on grayscale images are expected to be very limited.

Loading a textured 3D model. 3D models and their associated textures can be opened in OBJ or PLY format. For OBJ files, the texture must be referenced in the associated .mtl file. For PLY files, RASCAL looks for a texture image with the same base name as the PLY file, placed in the same folder. Once an image or a 3D model has been loaded, the full processing interface appears, the whitening transform is automatically computed.

Loading a previously recorded session. This file has recorded the full state of a previous session, including original image (or 3D model), current rotation, brush mask, selected parameters, and processing options (e.g. ICA). It can be reloaded later using:

File → Load session (or **Ctrl + Shift + L** / **⌘ + Shift + L** on macOS, or using drag and drop)

7. Processing an image

Image Quality (optional in 2D image processing). The *Image Quality Report* window is accessible from the menu:

File → Image Quality...

After calculation, a report appears in a window (Fig. 3). It provides a quick overview of the technical quality of the input image based on four parameters. For each metric, both the numerical value and a qualitative assessment are displayed in the report window:

Resolution (R): Evaluates the pixel count of the image relative to a 2-megapixel reference. Values above 1 indicate images exceeding the reference resolution. Labels: *sufficient resolution*, *borderline*, *low*, or *unusable*.

Quantization (Q): Assesses the JPEG quantization level, which reflects how aggressively the image was compressed at encoding time. Labels: *near lossless*, *light compression*, *moderate compression*, or *heavy compression*.

Blockiness (B): Detects the presence of 8×8 block artefacts typical of JPEG compression. These grid-like discontinuities become visible when an image has been saved at low quality or re-compressed multiple times. Labels: *no block artefacts*, *slight*, *visible*, or *strong*.

High frequency (HF): Measures the amount of fine detail preserved in the image by comparing its frequency spectrum to a Gaussian-blurred reference. A loss of high-frequency content indicates that the image has been smoothed, down-sampled, or heavily filtered. Labels: *very detailed*, *good*, *notable loss*, or *smoothed image*.

These four parameters are combined into a single SCI (Source Condition Index) score ranging from 0 to 100. The SCI is computed as the average of the four normalized components: Resolution (R), Quantization (Q), Blockiness (B), and High Frequency (HF), each contributing 25% to the final score. For the SCI calculation, the Resolution score is capped at its reference value, so that images exceeding the 2 Mpx reference do not receive additional weight. A score above 70 (displayed in green) indicates a well-preserved source; a score between 45 and 70 (orange) signals moderate degradation; a score below 45 (red) points to significant quality issues.

The Image Quality Report is provided as an indicative assessment of the input image's technical condition. It does not prejudge the results of the subsequent analysis. Images with lower scores may still yield meaningful results depending on the nature of the subject and the processing applied, while a high score does not guarantee that all features of interest will be successfully recovered. Users are encouraged to consider this report as a guideline rather than a definitive verdict.

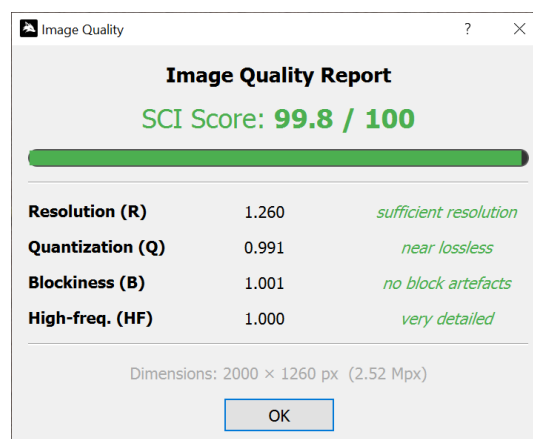


Figure 3: Screenshot of the Image Quality window. The SCI score (Source Condition Index) is computed from the resolution, quantization, blockiness, and high-frequency scores.

Modify Image (in all modes)

The *Modify Image* window (Figure 4) is accessible from the menu:

File → Modify Image...

It allows the user to modify the current image before it enters the RASCAL processing pipeline. Any applied modification triggers a full recomputation of the processing chain, including ZCA whitening, reset of the rotation state, and reconstruction of the 3D color distribution.

Cropping. To crop the image, draw a rectangle directly on the preview using the mouse (left click and drag). The selected area is displayed as a yellow dashed rectangle. If no selection is made, the entire image is retained.

Global image adjustments. The interface provides sliders for adjusting lightness, contrast, hue, saturation. All adjustments are applied in real time to the preview. The default value of each slider corresponds to no modification.

Individual RGB channel adjustments. Additional controls allow independent modification of the red, green and blue channels. These controls can be used to compensate for color imbalances or to emphasize subtle chromatic variations before RASCAL processing.

Resizing. The Width and Height fields (in pixels) allow resizing the resulting image. If a crop region is defined, the maximum dimensions correspond to the cropped area; otherwise, they correspond to the original image. The aspect ratio is preserved automatically since modifying one dimension updates the other accordingly. Resizing is performed using smooth bilinear interpolation.

Applying the modifications. The **OK** button applies all selected modifications (crop, color adjustments, and resizing) and reloads the resulting image into RASCAL. A complete recomputation of the processing pipeline is then performed. When the image has already been modified, a **Reset** button is displayed at the bottom of the window. It restores all adjustments to their initial values, clears the crop selection, restores the original dimensions, and immediately applies the reset to return to the original image state. Small × buttons are provided next to several controls to reset individual settings without closing the window. These resets update the preview immediately and allow individual modifications to be cancelled without affecting the other current settings.

TIP: Very dark images may contain insufficient separable chromatic information after the sRGB-to-linear RGB conversion. In such cases, applying a moderate increase in lightness before processing may improve the effectiveness of the whitening and rotation steps.

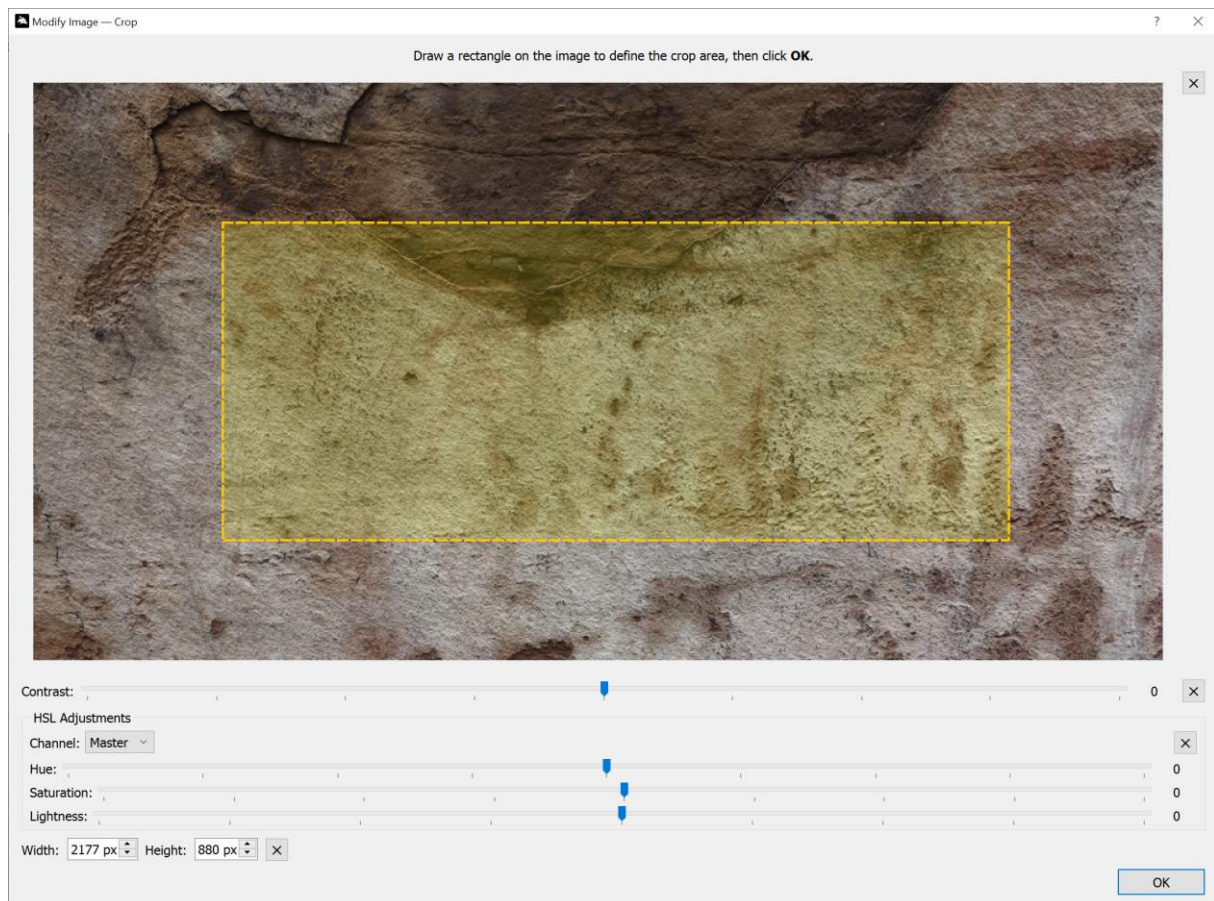


Figure 4: Screenshot of the **Modify Image** window. The interface provides tools for image cropping, resizing, and adjustment of global contrast, hue, saturation, and lightness, together with independent control of the red, green, and blue channels.

Interactive exploration (in all modes). Color transformations are explored interactively through direct mouse interaction on the preview canvas:

Left mouse drag (or keys **A / E**) → rotation around the Z axis

Right mouse drag (or keys **Z / S**) → rotation around the Y axis

Mouse wheel (or keys **Q / D**) → rotation around the X axis

Rotation → Auto rotate to perform automatic rotation. It is also possible to press the **Space bar**. Auto-rotation follows the most recently used rotation axis and direction.

Random → Random Rotation Explorer (only in Expert mode) launches an automatic sequence of random color-space orientations. A new random orientation is generated every 3 seconds, and the X, Y and Z values are updated accordingly. The sequence continues until the user interacts with any control, including a slider, mouse drag, or mouse wheel movement, at which point it stops immediately. Pressing Random again starts a new sequence.

Zooming can be performed using:

Ctrl + mouse wheel / ⌘ + mouse wheel on macOS
→ zoom in/out

Zooming is centered on the cursor position. When the zoom factor returns to 1×, the view is automatically recentered.

Rotation → Reset rotation resets the current rotation and zoom and returns the processing state to the current base image. It is also possible to press **R** or the corresponding toolbar icon: .

Region-of-interest (ROI) processing (in Expert mode). The user may optionally define a region of interest to guide the exploration. ROI mode can be activated using the **O** key or the **ROI** button in the interface. It involves manual selection of specific areas, from which the parameters for whitening transforms are extracted, and then applied to the entire image. Once activated, the user can paint a yellow region directly on the image (stroke width of the pen is configurable, 30 px by default). Pressing **Clear** deletes the selection. Pressing **Re-process** recomputes the transformation using only the selected pixels of the original image. This option allows color enhancement to focus on specific areas of interest while excluding external elements whose colors differ markedly from those of the region under investigation. Importantly, using a ROI modifies the statistical model of the image. As a result, transformations obtained with ROI processing are not directly comparable to those obtained from the full image. If activated, it is possible to deactivate ROI processing by pressing the **ROI** button twice. This resets the image and any complementary processing, while presets are preserved.

Independent component analysis (in Expert mode). ICA-assisted orientation is an optional automatic orientation of the whitened color space. It is computed after ZCA whitening (possibly with ROI) by pressing the **I** key or the **ICA** button, using a sample of up to 100,000 pixels. ICA does not replace manual exploration; it simply proposes an orientation of the whitened color space that may help separate statistically independent chromatic components. The user can then continue the exploration manually using rotation. If any modification occurs, the ICA button returns to white, indicating that the current orientation is no longer the one computed by ICA. ICA-assisted orientation and Radial Push are alternative optional processing modes in the current implementation. Activating Radial Push deactivates ICA, and ICA cannot be computed while Radial Push is active.

Polarity inversion (in Expert mode) The **Pol. +/-** button reverses the signs of the three Euler rotation angles used in the ZCA color space. In practice, the current rotation angles X, Y, and Z are replaced by -X, -Y, and -Z, and the rotation matrix is immediately recomputed from these new values. This provides a quick way to explore an alternative polarity of the current color enhancement, often making features that appear light on a dark background appear darker on a lighter background, or conversely. The displayed angle values, the processed image, and the point positions in the sRGB cube and ZCA space are updated accordingly. The operation is non-destructive and can be applied again to return to the previous angular configuration.

Radial push (in Expert mode). This function, activated using the **F** key or **Radial Push** button, expands the color distribution in decorrelated space. The method estimates the radial envelope of the color cloud using a spherical partition based on Fibonacci sampling. The resulting transformation reshapes the distribution of colors within the whitened space and may enhance color contrasts between the target features and the background. Importantly, the radial push is computed only once from the original whitened distribution and remains unchanged during subsequent rotations. The computation may take a few seconds depending on image size. Processing very high-resolution images in Radial Push mode on computers with limited memory may lead to a system crash. In such cases, RAM usage should be monitored; the information provided in the bottom right of the window. If necessary, images may be downsized prior to processing. Downscaling generally produces better results than applying strong JPEG compression, which is discouraged because it introduces visible blocking artifacts.

Push factor (in Expert mode). The push factor controls the global expansion of the color distribution in the whitened space. It acts as a multiplicative factor applied to the radial distance of all points from the origin while preserving their angular directions. Increasing the push factor expands the color cloud and enhances chromatic contrasts, whereas decreasing it reduces the overall amplitude of the transformation. The push factor can be adjusted using the **Push factor** control in the interface. It can also be modified using the keyboard: the **Up and Down arrow** keys increase or decrease the value. This parameter can be applied both in standard ZCA mode and after radial push normalization.

Split View (in Expert mode). The **Split** button divides a 2D image panel into two comparison areas. The left side displays the original reference image, while the right side shows the current processed rendering, including the active ZCA rotation, radial push, variance restoration, or channel display mode. A white vertical divider marks the boundary between the two views. The divider can be dragged horizontally with the mouse to adjust the relative size of the original and processed areas. When the image is zoomed or panned, the split remains aligned with the visible image area, making local before/after comparison easier. Split View can also be toggled quickly using the **T** keyboard shortcut.

Presets (in Expert mode). When exploring the color space, you may obtain a satisfying result but still wish to refine it further. Since navigating the space can make you lose a previous configuration, presets allow you to store intermediate checkpoints. This is particularly useful when working with multiple color patterns. Press **P** or use the **Presets** button to store the full processing state, including rotation parameters, push factor, radial push state, and any ROI-based reprocessing. Click on a preset number button to restore it. A maximum of 15 presets can be stored.

Variance Restore (in Expert mode). The Var slider controls how much of the original color variance/covariance structure is restored after whitening.

At 1.00 (default), the full inverse whitening transform is applied. The original variance structure is restored, producing the standard RASCAL rendering with more visually stable colors.

At 0.00, variance restoration is disabled. The decorrelated and variance-equalized color components are mapped directly for display, producing a stronger DStretch-like effect, with enhanced chromatic contrast and less natural colors.

Intermediate values provide a continuous transition between standard RASCAL rendering and a stronger decorrelation-stretch-like visualization. In practice, it is generally not necessary to decrease the Var slider below approximately 0.7-0.9.

Contrast / Auto Contrast. The Contrast slider adjusts the global contrast of the image manually. Moving the slider to the right increases contrast, while moving it to the left reduces it; the central position corresponds to no contrast modification. The **Auto Contrast** button automatically optimizes the image contrast by analyzing the luminance distribution and stretching it to use the available dynamic range more effectively, while preserving the overall color balance. This option can be particularly useful for dull images or when faint details are poorly visible.

Saving sessions (in all modes). Processing sessions can be saved in .rasc format using:

File → Session → Save session (or **Ctrl + Shift + S** / **⌘ + Shift + S** on macOS)

Session files store image or 3D model, image modifications (crop, resizing, brightness, contrast), rotation parameters, ROI selections, ICA, Radial Push and Push factor, and Variance Restore (but not presets) in a .rasc format. Sessions can later be restored using:

File → Session → Load session (or **Ctrl + Shift + L** / **⌘ + Shift + L** on macOS)

This functionality facilitates the reproducibility of image processing workflows when results are shared or published and is consistent with the FAIR principles for scientific data management. The Image Quality Report requires access to the original image file on disk. If the image was restored only from an embedded session file, this function may not be available.

Exporting results (in all modes). The current views can be exported using:

File → Export views (or the corresponding toolbar icon )

The user can choose which views to export by selecting the corresponding checkboxes. Available outputs include the processed color and Original (possibly modified with Modify Image) views, the R, G, and B channel views, and the 3D color cloud view. The exported color and R/G/B channel images are generated at the full resolution of the original image. They reproduce the current color transformation, including rotation, Radial Push, and variance restoration when enabled. The 3D color cloud is exported as a rendered visualization of the current display. A session file (.rasc) is automatically saved alongside the exported images.

All exported files follow a naming format: prefix_YYYYMMDD_HHMMSS_originalfilename_suffix.ext, where prefix is user-defined (default: export), YYYYMMDD_HHMMSS is the date and time of export, originalfilename is the name of the source image without extension, suffix is the view identifier, for

example color, R, G, B, 3d, original, and ext is the file extension, for example .jpg for images and .rasc for the session file.

Quitting the program (in all modes). Ending a session can be done using:

File → Quit (or **Ctrl + Q** / **⌘ + Q** on macOS)

If the session has not been saved, or if changes have been made since the last session save, a warning message will appear before quitting. The same warning is displayed when opening a new image or loading another session.

8. Processing a textured 3D model

Interactive exploration (in all modes). When working with a textured 3D model, RASCAL applies the color transformation to the texture image while simultaneously displaying the result on the 3D geometry. The interface is composed of two synchronized views: the texture image (left panel, Fig. 2A) and the rendered 3D model (right panel, Fig. 2B). In the 3D model window, mouse controls affect the visual orientation of the 3D geometry, not the chromatic rotation itself. The chromatic transformation is controlled from the texture view and by the keyboard shortcuts. In addition, panning of the 3D model view is available (**Ctrl + left mouse drag** / **⌘+ left mouse drag** on macOS).

Region-of-interest (ROI) processing (optional – only in Expert mode). As in standard image processing, selection can be performed directly on the texture. However, due to the fragmented nature of the UV map, this operation can be laborious when working on the 2D image. For this reason, painting directly on the 3D model is also possible, allowing more intuitive and precise targeting of meaningful areas.

Saving texture alone or model with modified texture. To save only the processed texture, use:

File → Export views,

while working in 3D mode. RASCAL exports the current processed texture as a JPG image. In 3D mode, Export views saves the processed texture using the `_rgb.jpg` suffix and automatically saves the associated .rasc session.

To save a complete model using the processed texture, use:

File → Save 3D model...,

which creates a new OBJ or PLY file and writes the transformed texture as a PNG file next to it. This command saves a new copy of the mesh in OBJ or PLY format and exports the processed texture as a PNG file next to it. The exported texture corresponds to the current RASCAL view, including the applied color transformation. The mesh geometry itself is not modified; only the texture image is replaced by the transformed one. Since the mesh is duplicated into a new file, exporting large models may require additional disk space.

Miscellaneous. Session saving, ICA, variance restore, and preset functions are available under the same conditions as for standard image processing. However, Radial Push is not available when processing a 3D model. Moreover, when a textured 3D model is being processed, Modify Image can be used to adjust the brightness and contrast of the texture, but cropping and resizing are disabled in order to preserve the texture-to-geometry mapping.

9. Quick reference

A brief recall of main controls appears at the start of a session as a pop-up window. It can be removed by checking the **‘Don’t show again’** box.

Action	Windows	macOS
Open a new file	Ctrl + O	⌘ + O
Rotate Z axis (image or 3D model)	Left mouse drag or A / E	Left drag or A / E
Rotate Y axis (image or 3D model)	Right mouse drag or Z / S	Right drag / two fingers click + drag or Z / S
Rotate X axis (image or 3D model)	Scroll wheel or Q / D	Scroll wheel / two-finger swipe or Q / D
Pan the 3D model view	Ctrl + left mouse drag	⌘ + left mouse drag
Zoom in/out	Ctrl + mouse wheel	⌘ + mouse wheel
Reset the view	R	R
Toggle 3D view (2D image processing)	N	N
ICA-assisted orientation	I	I
Split view	T	T
ROI mode	O	O
Radial Push	F	F
Presets	P	P
Increase push factor	Up arrow key	Up arrow key
Decrease push factor	Down arrow key	Down arrow key
Auto-rotate the whitened color space	Space bar	Space bar
Random rotation explorer	Ctrl + Shift + R	⌘ + Shift + R
Swap main view with channel view 1	Ctrl + 1	⌘ + 1

Action	Windows	macOS
Swap main view with channel view 2	Ctrl + 2	⌘ + 2
Swap main view with channel view 3	Ctrl + 3	⌘ + 3
Close a dialog box	ESC	ESC
Save session	Ctrl + Shift + S	⌘ + Shift + S
Load session	Ctrl + Shift + L	⌘ + Shift + L
Quit	Ctrl + Q	⌘ + Q

Reset and clearing commands. Several commands in RASCAL can either clear the current selection, exit ROI mode, or reset the processing state. They should not be confused, as they act at different levels. In the table below, the *base image* refers to the currently loaded image, including any modification previously applied through the Modify Image window, but before any ROI-based reprocessing.

Command	When to use it	Effect
Clear	During ROI painting	Clears only the current painted strokes. It does not modify the image, the current rotation, the ZCA model, ICA, Radial Push, Push Factor, Variance Restore, or saved presets.
Esc	While ROI painting mode is active	Exits ROI painting mode and clears the current unprocessed strokes. The current image and processing state are preserved.
Re-process	After painting a ROI	Recomputes the whitening model from the painted pixels and applies the resulting transformation to the whole image. The image is recomputed, the rotation is reset, ROI painting mode is exited, and the current brush is cleared.
ROI button pressed twice	When ROI mode is active	Deactivates ROI processing and returns the image to the current base image. Rotation, zoom, ICA, Radial Push, Push Factor, Variance Restore, and brush information are reset. Presets are preserved.
Reset / R	To return to the current base image	Performs a broader reset: the image returns to the current base image, the whitening model is recomputed, rotation and zoom are reset, ICA and Radial Push are deactivated, Push Factor and Variance Restore return to their default values, brush/ROI information is cleared, and the default view arrangement is restored. Presets are preserved.
Modify Image → OK	After cropping, resizing, or adjusting brightness / contrast	The modified image becomes the new base image. The whitening model is recomputed from this modified image, and rotation, zoom, ROI, ICA, Radial Push, Push Factor, Variance Restore, and presets are reset.

10. Limitations

RASCAL continuously monitors memory usage and displays the current value in the status bar, at the bottom right of the main window. When the overall system memory usage exceeds 80% of the available RAM, the indicator turns red to warn the user. Before potentially memory-intensive operations, such as loading a large image, enabling Radial Push, saving a preset, or reprocessing a ROI, the application checks the current memory status and may ask the user for confirmation if the warning threshold has already been reached. When opening an image larger than 30 megapixels, RASCAL also displays an estimate of the expected memory footprint, including the additional memory required for each preset checkpoint. If problems or crashes occur, downscaling the image prior to processing is recommended.

The method assumes a relatively continuous color distribution. When multiple distinct color populations are present, ROI-based processing can be used; however, it may bias the transformation toward the selected region. In such cases, the operator is encouraged to explore separate configurations for each color population of interest. When high-bit-depth or floating-point images are embedded in .rasc session files, they are stored as 8-bit PNG images. The session restores the visual state of the processing, but numerical precision beyond 8 bits is not preserved in the embedded images.

11. Final note

The software is provided without warranty regarding the interpretation of results. Image transformations should always be interpreted within the archaeological or scientific context of the studied material.

12. Contact

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